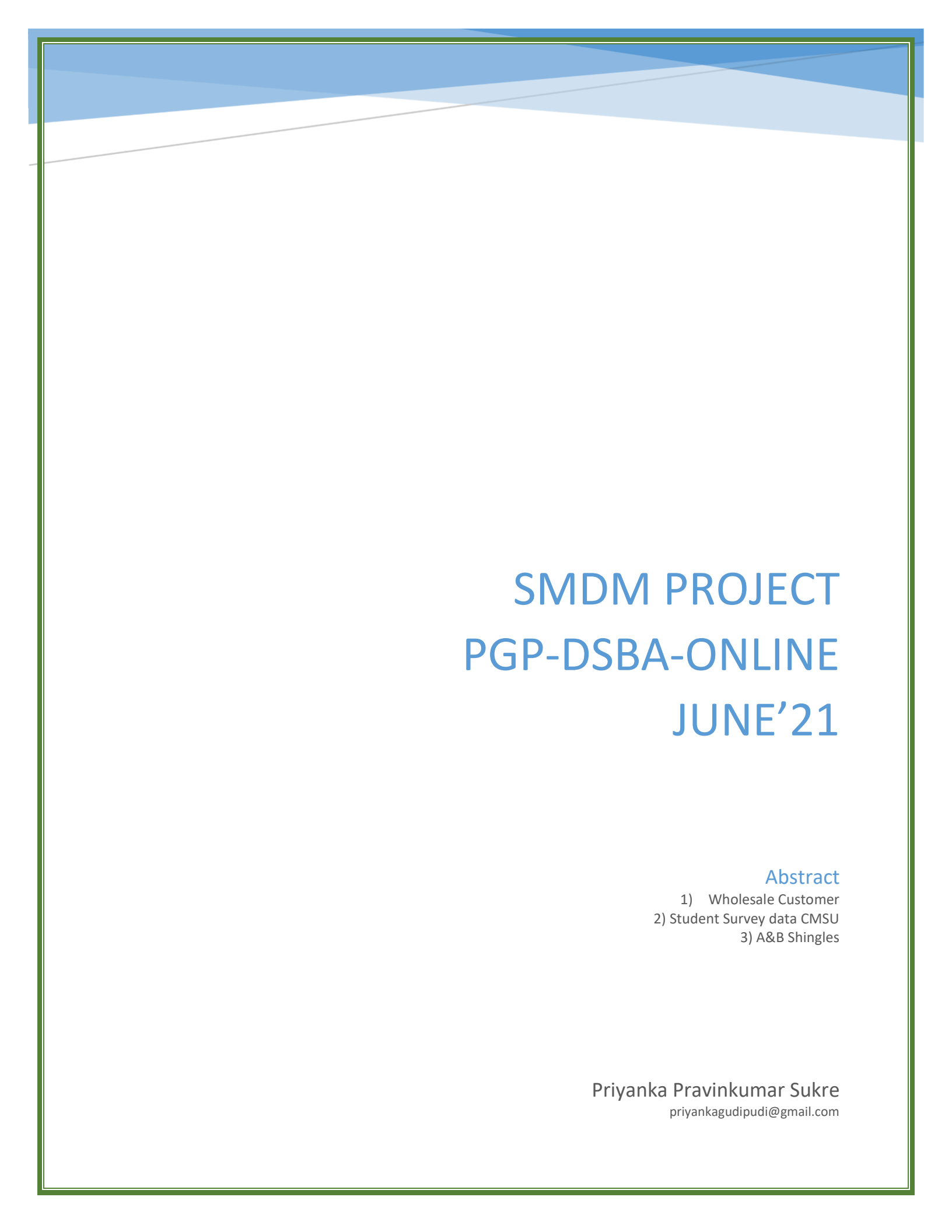




SMDM PROJECT PGP-DSBA-ONLINE JUNE'21

Abstract

- 1) Wholesale Customer
- 2) Student Survey data CMSU
- 3) A&B Shingles

Priyanka Pravinkumar Sukre
priyankagudipudi@gmail.com

Table of Contents

Problem Statement: Whole Sale Customer	3
Introduction:	3
Sample of the dataset:	3
Exploratory data analysis:	4
1.1 Use methods of descriptive statistics to summarize data. Which Region and which Channel spent the most? Which Region and which Channel spent the least?	4
1.2 There are 6 different varieties of items that are considered. Describe and comment/explain all the varieties across Region and Channel? Provide a detailed justification for your answer.	5
1.3 On the basis of a descriptive measure of variability, which item shows the most inconsistent behavior? Which items show the least inconsistent behavior?	7
1.4 Are there any outliers in the data? Back up your answer with a suitable plot/technique with the help of detailed comments.	7
1.5 On the basis of your analysis, what are your recommendations for the business? How can your analysis help the business to solve its problem? Answer from the business perspective.	8
Problem Statement: Student Survey Data	9
Introduction:	Error! Bookmark not defined.
Sample of the dataset:	9
2.1. For this data, construct the following contingency tables (Keep Gender as row variable)	10
2.2. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:	11
2.3. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:	11
2.4. Assume that the sample is a representative of the population of CMSU. Based on the data, answer the following question:	13
2.5. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:	13
2.6. Construct a contingency table of Gender and Intent to Graduate at 2 levels (Yes/No). The Undecided students are not considered now and the table is a 2x2 table. Do you think the graduate intention and being female are independent events?	14

2.7. Note that there are four numerical (continuous) variables in the data set, GPA, Salary, Spending, and Text Messages.	15
2.8. Note that there are four numerical (continuous) variables in the data set, GPA, Salary, Spending, and Text Messages. For each of them comment whether they follow a normal distribution. Write a note summarizing your conclusions for this whole Problem 2.	16
2.8.2 Write a note summarizing your conclusions for this whole Problem?	17
Problem Statement: A & B Asphalt Shingles	18
Introduction:	18
3.1 Do you think there is evidence that means moisture contents in both types of shingles are within the permissible limits? State your conclusions clearly showing all steps.....	18
3.2 Do you think that the population mean for shingles A and B are equal? Form the hypothesis and conduct the test of the hypothesis. What assumption do you need to check before the test for equality of means is performed?	20

PROJECT 1: Whole Sale Customer

Problem Statement: Whole Sale Customer

A wholesale distributor operating in different regions of Portugal has information on annual spending of several items in their stores across different regions and channels. The data consists of 440 large retailers' annual spending on 6 different varieties of products in 3 different regions (Lisbon, Oporto, Other) and across different sales channel (Hotel, Retail).

Introduction:

The purpose of this assignment to perform the data analysis for understanding and analyzing the data to make best possible decisions for business to perform. The data consists of 6 different varieties of items across 2 sales channels (Retail and Hotel) in 3 different regions (Lisbon, Oporto and other). This analysis will help the Wholesale distribute understand the spending behavior in these regions and act accordingly in distribution of items to these places.

Sample of the dataset:

	Buyer/Spender	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicatessen
0	1	Retail	Other	12669	9656	7561	214	2674	1338
1	2	Retail	Other	7057	9810	9568	1762	3293	1776
2	3	Retail	Other	6353	8808	7684	2405	3516	7844
3	4	Hotel	Other	13265	1196	4221	6404	507	1788
4	5	Retail	Other	22615	5410	7198	3915	1777	5185
6	6	Retail	Other	9413	8259	5126	666	1795	1451
6	7	Retail	Other	12126	3199	6975	480	3140	545
7	8	Retail	Other	7579	4956	9426	1669	3321	2566
8	9	Hotel	Other	5963	3648	6192	425	1716	750
9	10	Retail	Other	6006	11093	18881	1159	7425	2098

Table 1. Dataset of the Sample

From the above dataset it is evident that there are 6 different items across three different regions and two sales channels from which the spending behavior can be analyzed.

Exploratory data analysis:

RangeIndex: 440 entries, 0 to 439

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Buyer/Spender	440 non-null	int64
1	Channel	440 non-null	object
2	Region	440 non-null	object
3	Fresh	440 non-null	int64
4	Milk	440 non-null	int64
5	Grocery	440 non-null	int64
6	Frozen	440 non-null	int64
7	Detergents_Paper	440 non-null	int64
8	Delicatessen	440 non-null	int64

From the above information, it is clear that the dataset is having 440 entries with string and integer data types.

1.1 Use methods of descriptive statistics to summarize data. Which Region and which Channel spent the most? Which Region and which Channel spent the least?

Summary of the dataset is as follows:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Buyer/Spender	440	NaN	NaN	NaN	220.5	127.161	1	110.75	220.5	330.25	440
Channel	440	2	Hotel	298	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Region	440	3	Other	316	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Fresh	440	NaN	NaN	NaN	12000.3	12647.3	3	3127.75	8504	16933.8	112151
Milk	440	NaN	NaN	NaN	5796.27	7380.38	55	1533	3627	7190.25	73498
Grocery	440	NaN	NaN	NaN	7951.28	9503.16	3	2153	4755.5	10655.8	92780
Frozen	440	NaN	NaN	NaN	3071.93	4854.67	25	742.25	1526	3554.25	60869
Detergents_Paper	440	NaN	NaN	NaN	2881.49	4767.85	3	256.75	816.5	3922	40827
Delicatessen	440	NaN	NaN	NaN	1524.87	2820.11	3	408.25	965.5	1820.25	47943

Table 2 : Descriptive Statistics

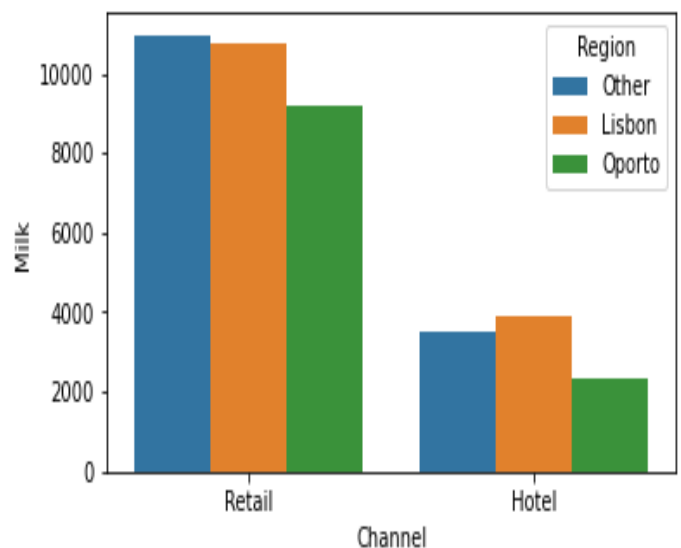
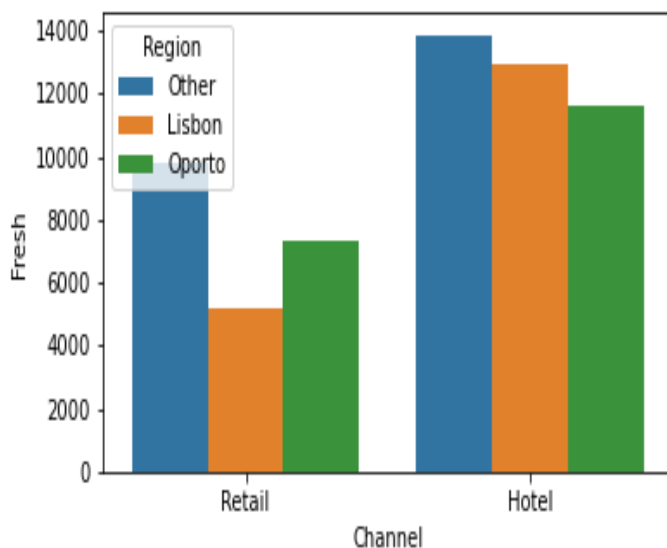
From the above description, it is clearly seen that the Other Region and Hotel Channel have more annual spending of various items than the other channels and Regions.

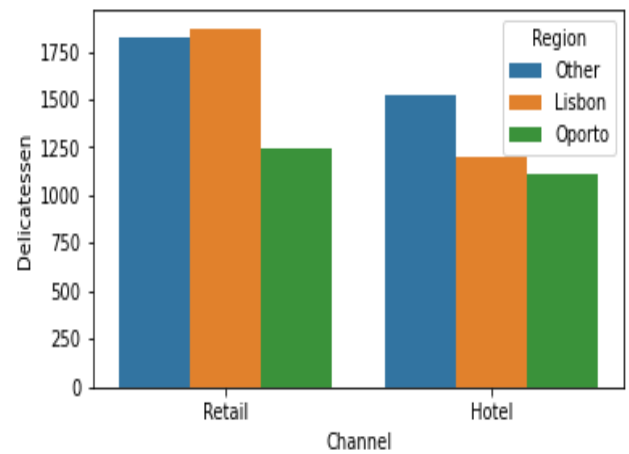
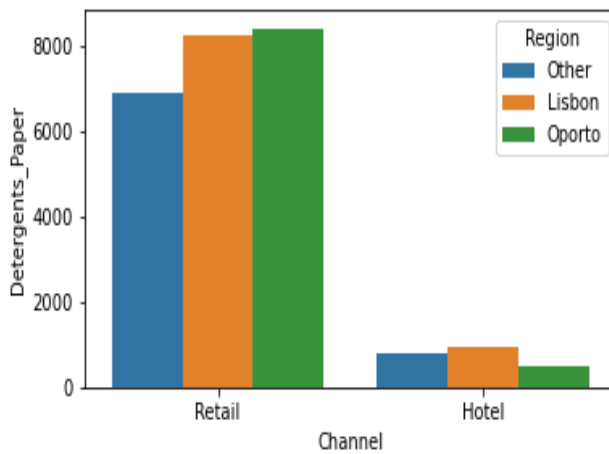
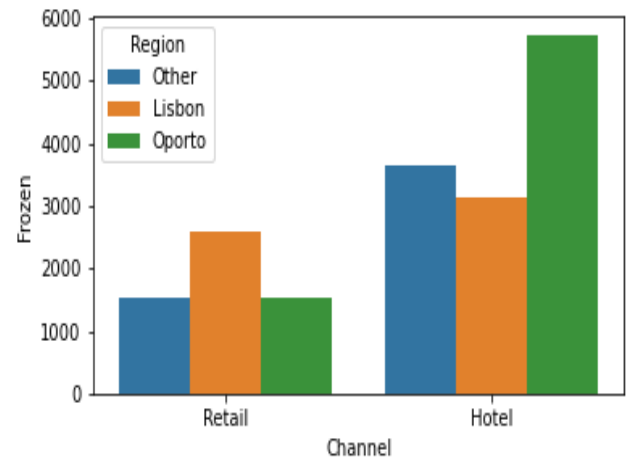
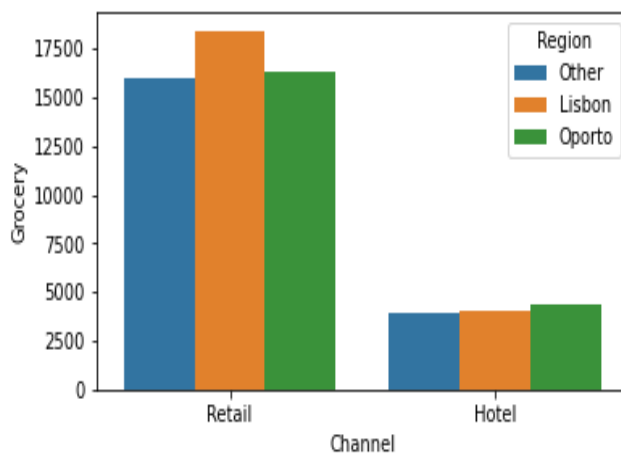
Below is the table showing the Annual spending of different items across regions and Channels.

Region	Lisbon	Oporto	Other
Channel			
Hotel	1538342	719150	5742077
Retail	848471	835938	4935522

- 1) Hotel chain in Other region spent the most of about 57,42,077
- 2) Hotel chain in Porto Region spent the least of about Rs. 7,19,150

1.2 There are 6 different varieties of items that are considered. Describe and comment/explain all the varieties across Region and Channel? Provide a detailed justification for your answer.





Summarizing the graphical representation in to the Tabular form as below:

S No	Items	Highest Spending	Lowest Spending
1)	Fresh	Hotel Channel in Other Region	Retail Chain in Lisbon Region
2)	Milk	Retail Chain in Other Region	Hotel Chain in Oporto Region
3)	Grocery	Retail Chain in Lisbon Region	Hotel Chain in Other Region
4)	Frozen	Hotel Chain in Oporto Region	Retail Chain in Other Region
5)	Detergents_Paper	Retail Chain in Oporto Region	Hotel Chain in Oporto Region
6)	Delicatessen	Retail Chain in Lisbon Region	Retail Chain in Oporto Region

We can evidently see that, it is evident that Milk, Grocery and Detergent Papers have highest spending in Retail Channel in all regions while fresh and frozen have highest sensing in Hotel Channel. Here the Wholesale Distributor can focus on the items based on the highest spending channels in the region. For example: Delicatessen in retail channel have high spending in Lisbon

region while lowest spending in Oporto Region. Basis this, the distributor can plan items distributions accordingly.

1.3 On the basis of a descriptive measure of variability, which item shows the most inconsistent behavior? Which items show the least inconsistent behavior?

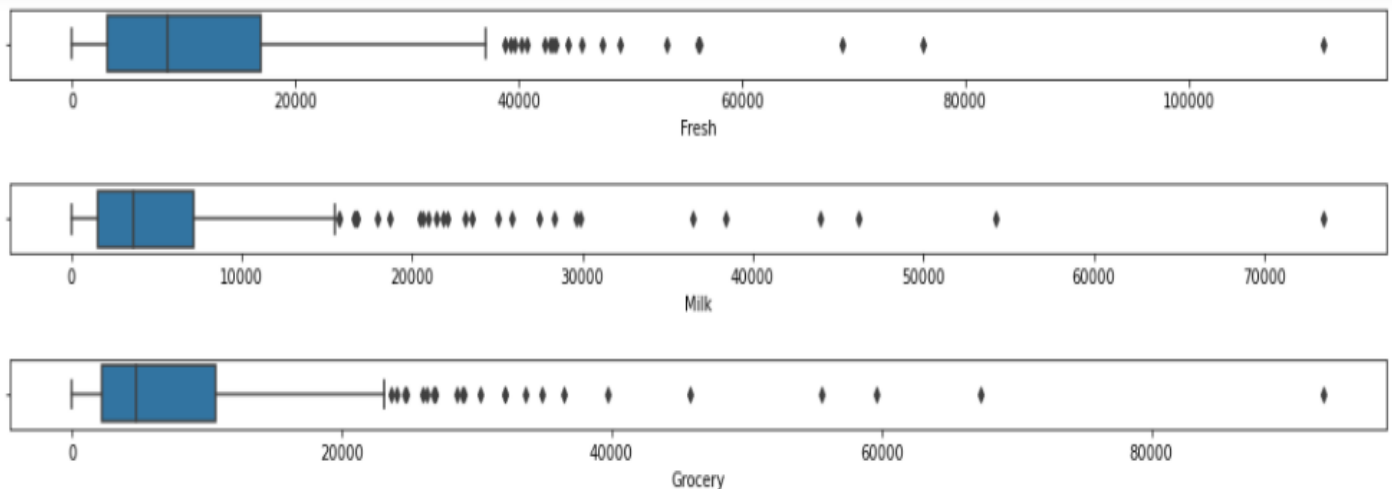
Items	Co-efficient of Variation
Buyer/Spender	0.576695
Fresh	1.053918
Milk	1.273299
Grocery	1.195174
Frozen	1.580332
Detergents_Paper	1.654647
Delicatessen	1.849407

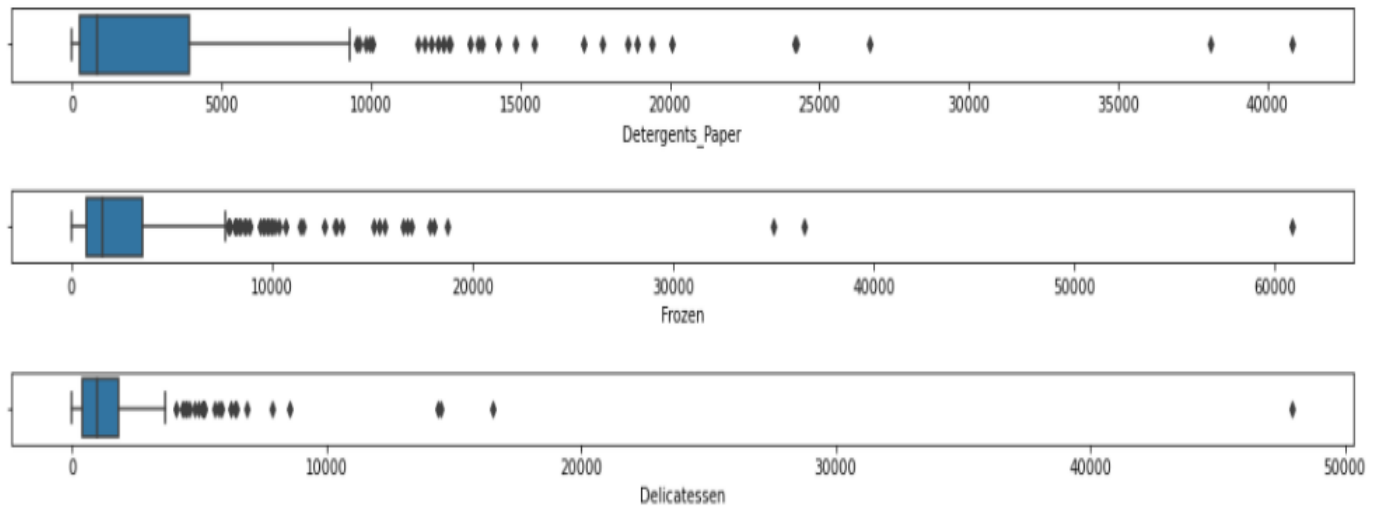
Name: Co-efficeint of Variation,

From the above consistency tables, it is evident that, Delicatessen having the most inconsistent behavior and Fresh is having the least Inconsistent behavior, therefore, the distributor should maintain the consistency in distributing the Fresh item.

1.4 Are there any outliers in the data? Back up your answer with a suitable plot/technique with the help of detailed comments.

A Box plot is the good representation to show if there are any outliers in the dataset.





From the above graphical representation it is evident that, Outliers exists in all of the items present in dataset and it is clear that Fresh and Grocery have highest spending across regions and channels while Delicatessen have the least.

1.5 On the basis of your analysis, what are your recommendations for the business? How can your analysis help the business to solve its problem? Answer from the business perspective.

As different items have different behavior in spending across the all regions and channels, the distribution in Fresh Item needs to be maintained consistently. Distribution can be planned accordingly based on the highest spending in the regions and channels so the cost of maintaining stock may be minimized. Procurement cost for the items distribution will be made easy as we know that which item is performing well in which area.

PROJECT 2: Student Survey Data

Problem Statement: Student Survey Data

The Student News Service at Clear Mountain State University (CMSU) has decided to gather data about the undergraduate students that attend CMSU. CMSU creates and distributes a survey of 14 questions and receives responses from 62 undergraduates (stored in the **Survey** data set).

Sample of the dataset:

	ID	Gender	Age	Class	Major	Grad Intention	GPA	Employment	Salary	Social Networking	Satisfaction	Spending	Computer	Text Messages
0	1	Female	20	Junior	Other	Yes	2.9	Full-Time	50.0	1	3	350	Laptop	200
1	2	Male	23	Senior	Management	Yes	3.6	Part-Time	25.0	1	4	360	Laptop	50
2	3	Male	21	Junior	Other	Yes	2.5	Part-Time	45.0	2	4	600	Laptop	200
3	4	Male	21	Junior	CIS	Yes	2.5	Full-Time	40.0	4	6	600	Laptop	250
4	5	Male	23	Senior	Other	Undecided	2.8	Unemployed	40.0	2	4	500	Laptop	100

Table 1. Sample Dataset

From the above sample dataset, we can conclude that the student data is

2.1. For this data, construct the following contingency tables (Keep Gender as row variable)

2.1.1. Gender and Major

Major	Accounting	CIS	Economics/Finance	International Business	Management	Other	Retailing/Marketing	Undecided
Gender								
Female	3	3	7	4	4	3	9	0
Male	4	1	4	2	6	4	5	3

From the above contingency table we can see that, More number of Female opted for Retailing and more number of Male opted for Management while less male opted for CIS major.

2.1.2. Gender and Grad Intention

Grad Intention	No	Undecided	Yes
Gender			
Female	9	13	11
Male	3	9	17

From the table it is clear that, Males have more Grad Intention than Females.

2.1.3. Gender and Employment

Employment	Full-Time	Part-Time	Unemployed
Gender			
Female	3	24	6
Male	7	19	3

More number of students both Female and Male have part-time employment and Female candidates are more unemployed than male candidates.

2.1.4. Gender and Computer

	Computer	Desktop	Laptop	Tablet
Gender				
Female		2	29	2
Male		3	26	0

More number of students both Male and Female prefer laptops than desktop and tablet.

2.2. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:

From the data we can get the Total Number of students Gender wise as below:

```
Female    33
Male      29
Name: Gender, dtype: int64
```

Both Female and Male accounts for 62 students.

2.2.1. What is the probability that a randomly selected CMSU student will be male?

The probability that a randomly chosen student will be a male is: **46.77%**

2.2.2. What is the probability that a randomly selected CMSU student will be female?

The probability that a randomly chosen student will be a female is: **53.23%**

2.3. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:

2.3.1. Find the conditional probability of different majors among the male students in CMSU.

From the Contingency Table we can have the below table:

Major	Accounting	CIS	Economics/Finance	International Business	Management	Other	Retailing/Marketing	Undecided
Gender								
Female	3	3	7	4	4	3	9	0
Male	4	1	4	2	6	4	5	3

So,

Probability of a male student opting for Accounting Major is **13.79%**

Probability of a male student opting for CIS Major is **3.45%**

Probability of a male student opting for Economics/finance Major is **13.79%**

Probability of a male student opting for International Business Major is **6.90%**

Probability of a male student opting for Management Major is **20.69%**

Probability of a male student opting for Other Major is **13.79%**

Probability of a male student opting for Retailing/Marketing Major is **17.24%**

Probability of a male student opting for Undecided Major is **10.34%**

2.3.2 Find the conditional probability of different majors among the female students of CMSU.

From the above contingency table,

Probability of a Female student opting for Accounting Major is **9.09%**

Probability of a Female student opting for CIS Major is **9.09%**

Probability of a Female student opting for Economics/finance Major is **21.21%**

Probability of a Female student opting for International Business Major is **12.12%**

Probability of a Female student opting for Management Major is **12.12%**

Probability of a Female student opting for Other Major is **9.09%**

Probability of a Female student opting for Retailing/Marketing Major is **27.27%**

Probability of a Female student opting for Undecided Major is **0.00%**

2.4. Assume that the sample is a representative of the population of CMSU. Based on the data, answer the following question:

2.4.1. Find the probability that a randomly chosen student is a male and intends to graduate.

Using the Contingency tables of Gender and Grad Intention, we go the below table

Grad Intention	No	Undecided	Yes
Gender			
Female	9	13	11
Male	3	9	17

From the table, we can say that, the probability that a randomly chosen student is Male and intends to graduate is **58.62%**

2.4.2 Find the probability that a randomly selected student is a female and does NOT have a laptop.

Using the Contingency tables of Gender and Computer, we go the below table,

Computer	Desktop	Laptop	Tablet
Gender			
Female	2	29	2
Male	3	26	0

From the table, we can say that, the probability that a randomly chosen student is a Female and does not have laptop is **12.12%**

2.5. Assume that the sample is representative of the population of CMSU. Based on the data, answer the following question:

2.5.1. Find the probability that a randomly chosen student is a male or has full-time employment?

Using the Contingency tables of Gender and Employment, we get the total number of male having employment,

Employment	Full-Time	Part-Time	Unemployed
Gender			
Female	3	24	6
Male	7	19	3

So, the probability that a randomly chosen student is a male or has a full-time employment is **70.91%**

2.5.2. Find the conditional probability that given a female student is randomly chosen, she is majoring in international business or management.

Using the Contingency tables of Gender and Major, we get the total number of female majoring in international business or management,

So the, conditional probability that given a female student is randomly chosen, she is majoring in international business or management. **24.24%**

2.6. Construct a contingency table of Gender and Intent to Graduate at 2 levels (Yes/No). The Undecided students are not considered now and the table is a 2x2 table. Do you think the graduate intention and being female are independent events?

Using the contingency table and dropping the undecided column we get the following table:

Grad Intention	No	Yes	Total
Gender			
Female	9	11	20
Male	3	17	20
Grand Total	12	28	40

The Probability of Graduate Intention is = $28/40 = 0.7$

The probability of Graduate Intention and being a female candidate is = $11/20 = 0.55$

So, the graduate intention and being female are independent events.

2.7. Note that there are four numerical (continuous) variables in the data set, GPA, Salary, Spending, and Text Messages.

Answer the following questions based on the data

2.7.1. If a student is chosen randomly, what is the probability that his/her GPA is less than 3?

The Gender wise table of his/her GPA is less than 3 is as below:

```
Male      9
Female    8
Name: Gender, dtype: int64
```

If a student is chosen randomly, the probability that his/her GPA is less than 3 is **27.42%**

**2.7.2. Find the conditional probability that a randomly selected male earns 50 or more.
Find the conditional probability that a randomly selected female earns 50 or more.**

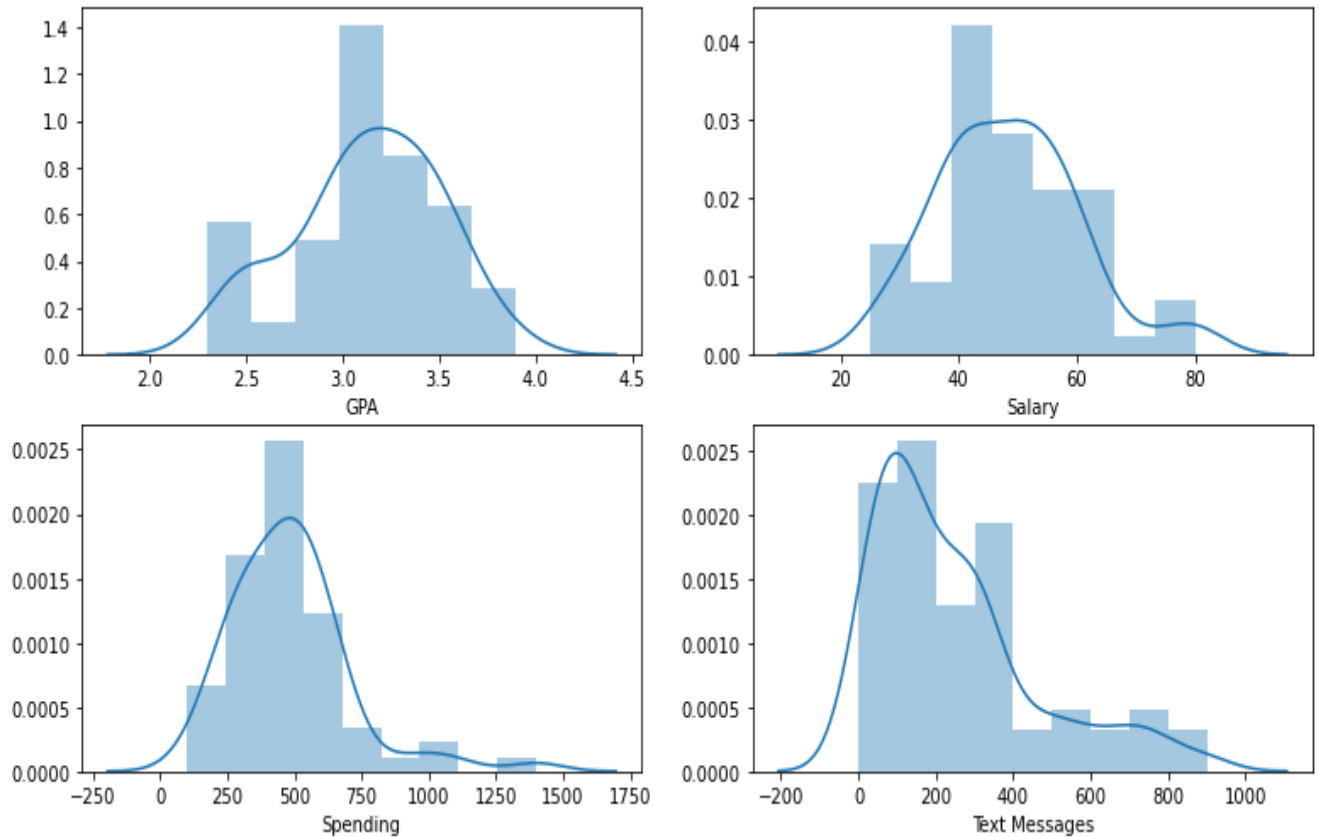
The Gender wise table of earnings of 50 or more is as below:

```
Female    18
Male      14
Name: Gender, dtype: int64
```

The Probability that a randomly selected male earns 50 or more is **48.28%**

The Probability that a randomly selected female earns 50 or more is **54.55%**

2.8. Note that there are four numerical (continuous) variables in the data set, GPA, Salary, Spending, and Text Messages. For each of them comment whether they follow a normal distribution. Write a note summarizing your conclusions for this whole Problem 2.



GPA:

- ShapiroResult(statistic=0.9685361981391907, pvalue=0.11204058676958084)
- Level of significance: 0.05
- We have no evidence to reject the null hypothesis since p value > Level of significance so the data is normally distributed

Salary:

- ShapiroResult(statistic=0.9565856456756592, pvalue=0.028000956401228905)
- Level of significance: 0.05

- We have evidence to reject the null hypothesis since $p \text{ value} < \text{Level of significance}$ so data is not normally distributed

Spending:

- ShapiroResult(statistic=0.8777452111244202, pvalue=1.6854661225806922e-05)
- Level of significance: 0.05
- We have evidence to reject the null hypothesis since $p \text{ value} < \text{Level of significance}$ so data is not normally distributed

Text Messages:

- ShapiroResult(statistic=0.8594191074371338, pvalue=4.324040673964191e-06)
- Level of significance: 0.05
- We have evidence to reject the null hypothesis since $p \text{ value} < \text{Level of significance}$ so data is not normally distributed

2.8.2 Write a note summarizing your conclusions for this whole Problem?

Here, the analysis done on the male and female student observations across majors, employment, earnings, etc., From this the major observation is, though the probability of the being a female candidate in the dataset is high but the probability of male getting high employment is more than female. Here we also can observe that most female prefer Major in Retail/Marketing while most male prefer in major in management.

Among the four continuous variable only one variable is normally distributed. An increase in sample in sample size may be done to make the variable normally distributed so any further hypothesis may be conducted in the problem.

PROJECT 3: A & B Asphalt Shingles

Problem Statement: A & B Asphalt Shingles

An important quality characteristic used by the manufacturers of ABC asphalt shingles is the amount of moisture the shingles contain when they are packaged. Customers may feel that they have purchased a product lacking in quality if they find moisture and wet shingles inside the packaging. In some cases, excessive moisture can cause the granules attached to the shingles for texture and colouring purposes to fall off the shingles resulting in appearance problems. To monitor the amount of moisture present, the company conducts moisture tests. A shingle is weighed and then dried. The shingle is then reweighed, and based on the amount of moisture taken out of the product, the pounds of moisture per 100 square feet is calculated. The company would like to show that the mean moisture content is less than 0.35 pound per 100 square feet.

Introduction:

Customers of the manufacturers of ABC asphalt have felt that the moisture content for the products quality is lacking. To test this company wants to check whether the moisture content is less than 0.35 pound per 100 square feet. For this test to be conducted we need to perform the hypothesis testing of null hypothesis and alternate hypothesis.

3.1 Do you think there is evidence that means moisture contents in both types of shingles are within the permissible limits? State your conclusions clearly showing all steps.

The permissible limit of the moisture is less than **0.35 pound**

Hence, for every moisture test, the claim to check the whether the moisture content is still greater than 0.35 pounds per 100 square feet.

Alternative hypothesis (H_A) : mean moisture content > 0.35

Null hypothesis (H₀) : mean moisture content ≤ 0.35

Here the shingles of A and B will be performed separately by conducting T test hypothesis of one tailed sample.

Hypothesis testing for A shingle:

Null Hypothesis H_0 : Mean Moisture ≤ 0.35

Alternate Hypothesis H_1 : Mean Moisture > 0.35

Here, the test is right sided test. We have to find out the T_{stat} value and p_{value} whether to accept or reject the null hypothesis.

By performing the one sample T Test we get t_{stat} and p_{value} :

t_{stat} value = -1.4735046253382782

p value = 0.07477633144907513

Level of significance: 0.05

We have no evidence to reject the null hypothesis since p value $>$ Level of significance

Our one-sample $t_{\text{stat_value}}$ $p = 0.07477633144907513$

From the above output we can say that the mean moisture for A shingles have in permissible limits i.e, less than 0.35 moisture content per 100 sq.ft

Hypothesis testing for B shingle:

Null Hypothesis H_0 : Mean Moisture ≤ 0.35

Alternate Hypothesis H_1 : Mean Moisture > 0.35

Here, the test is right sided test. We have to find out the T_{stat} value and p_{value} whether to accept or reject the null hypothesis.

By performing the one sample T Test we get t_{stat} and p_{value} :

t_{stat} value = -3.1003313069986995

p value = 0.0020904774003191826

Level of significance: 0.05

We have evidence to reject the null hypothesis since p value $<$ Level of significance

Our one-sample $B_{\text{tstat_value}}$ $B_P = 0.0020904774003191826$

From the above output we can say that the mean moisture for B shingles have is not in permissible limits i.e, the moisture content is greater than, 0.35 pound per 100 square feet.

3.2 Do you think that the population mean for shingles A and B are equal? Form the hypothesis and conduct the test of the hypothesis. What assumption do you need to check before the test for equality of means is performed?

Assumption: Both the samples are normally distributed and the variance of both the samples are equal. If either of one is not true we may need to perform so other test to meet the assumptions of normal distribution and equal variance.

The population mean for shingles A and B are not equal.

Two sample T test is as follows.

Null Hypothesis H_0 : Mean Moisture ≤ 0.35

Alternate Hypothesis H_1 : Mean Moisture is not equal to 0.35

Here, The test is two sided test.

tstat_value = 1.2896282719661123

P_value = 0.2017496571835306

Level of significance: 0.05

We have no evidence to reject the null hypothesis since p value > Level of significance

Our two-sample tstat_value p_value= 0.2017496571835306

From the above conclusion we can say that, the mean moisture for the A and B shingles are with in the permissible limit of 0.35 pound per square feet.